

SOFTWARE TESTING

MASTERCLASS

Types of testing Black-box and White-box

Black Box testing is a software testing method in which the internal implementation of the function is not known by the tester. This means that the tester doesn't know how the application's "code" works

Incorrect
or missing
functions

Interface
/ Visual
Errors

Behavior
issues

Initialization
or
termination
errors





Basics of Software Testing

Types of testing Black-box and White-box

- **Functional testing** - This Black Box testing type is related to the functional requirements of a system.
- **Non-functional testing** - This is a type of Software testing that checks non-functional aspects of a software application.
- **Regression testing** involves testing existing software applications to ensure that a change or addition hasn't broken any existing functionality.





Basics of Software Testing

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Functional Testing	Non - Functional Testing
1- Tester tests how well the system performs.	1- Tester tests how well the system responds
2- Based on client requirements.	2- Based on client expectations.
3- Testing the application as per the business requirements.	3- Testing the application against clients and performance requirements.
4- Part of System Testing	4- Part of System Testing
5- Validates the behavior of an application	5- Validates the performance of an application
6- Covers Unit Testing, Integration Testing, Smoke Testing, Sanity Testing, Regression Testing	6- Covers Load/Performance Testing, Stress/Volume Testing, Security Testing, Installation Testing
7- Concentrates on customer requirements.	7- Concentrates on customer expectations.
8- How is your system doing.	8- How well your system is doing





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Basics of Software Testing

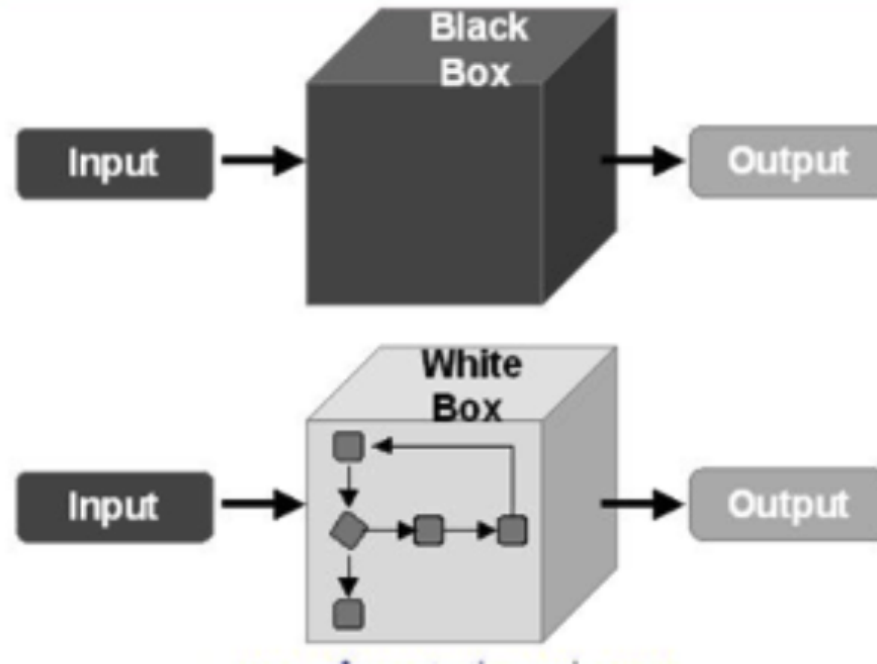
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- **Equivalence Class Testing:** A Black Box test technique in which test cases are designed to exercise one representative member of each partition.
- **Boundary Value Testing:** This technique focuses on the values at boundaries. It determines whether a certain range of values is acceptable by the system or not.
- **Decision Table Testing:** This is a visual representation of which actions to perform depending on given conditions. It creates all possible decisions for the user.



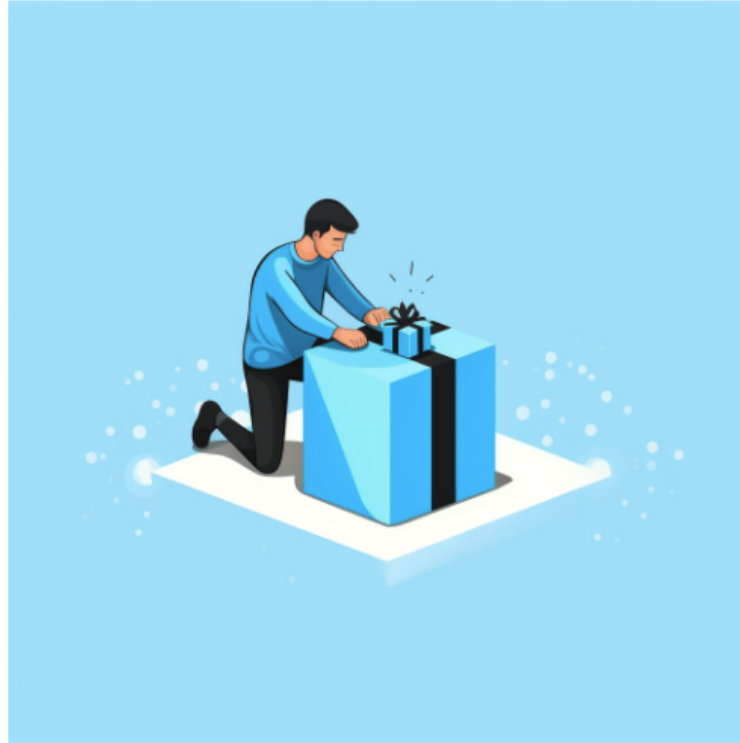
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Black Box Techniques:

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table State
- Transition Testing
- Use Case Testing

Some White Box techniques are:

- Unit Testing
- Statement Testing

